

MotoTrack Web-Based Management System

1. Main System Flow

Customer Flow

1. Customer visits the MotoTrack website.
2. Customer creates an account or logs in.
3. Customer selects motorcycle information first:
 - Motorcycle type:
 - Underbone
 - Backbone
 - Scooter
 - Brand
 - Model
Example: Honda Click
4. Based on the selected motorcycle type, the system automatically displays compatible services.

Example:

- If the customer selects **Scooter**, the system will show:
 - CVT Cleaning
 - Change Oil
 - General Maintenance
- If the customer selects **Underbone** or **Backbone**, the system will not show CVT Cleaning because these motorcycles do not have CVT parts. Instead, the system will show:
 - Chain Cleaning
 - Sprocket Cleaning
 - Change Oil
 - General Maintenance

5. Customer selects the desired maintenance or service.
6. The system asks follow-up questions depending on the selected service.

Example for Change Oil:

- Customer selects motorcycle engine size:
 - 110cc to 125cc
 - 150cc and above

System suggestion:

- 110cc to 125cc motorcycles usually need **0.8 liter** of engine oil.
- 150cc and above motorcycles usually need **1 liter** of engine oil.

Example for CVT Cleaning:

- 125cc below: system suggests **2 cans of CVT cleaner**.
 - 150cc and above: system suggests **3 cans of CVT cleaner**.
7. The system automatically calculates the estimated service cost.

Example:

- Change Oil Service:
 - Labor fee: ₱50
 - Plus engine oil price
- CVT Cleaning:
 - CVT cleaner cost
 - Labor fee: ₱300

8. Customer checks:

- Product price
- Product availability
- Estimated service cost
- Available appointment schedule

9. Customer can:

- Reserve motorcycle/unit
 - Reserve parts for pickup
 - Book repair or maintenance appointment
10. System records the reservation or appointment.
 11. Staff receives a notification.
 12. Staff confirms, updates, or cancels the reservation.
 13. Customer receives reservation status update.
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2. Staff / Cashier Flow

1. Staff logs in.
 2. Staff opens the POS module.
 3. Staff searches or scans an item.
 4. System checks stock availability.
 5. Staff adds item to cart.
 6. System calculates subtotal and total.
 7. Staff processes payment.
 8. System automatically deducts inventory.
 9. Receipt or invoice is generated.
 10. Sale is saved for reporting.
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3. Technician Flow

1. Technician logs in.
2. Technician views service appointments.
3. Technician checks customer motorcycle details and service request.
4. Technician updates service status:
 - Pending

- In Progress
 - Completed
 - Cancelled
5. Technician records used parts and materials.
 6. Used parts are deducted from inventory.
 7. Final service cost is saved.
 8. Service record is stored in the system.
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4. Admin / Owner Flow

1. Admin logs in.
2. Admin manages:
 - Users
 - Products
 - Inventory
 - Services
 - Reservations
 - Sales
 - Reports
 - Shop settings
3. Admin views:
 - Daily sales
 - Monthly sales
 - Low-stock alerts
 - Inventory turnover
 - Reservation history
 - Service appointment reports

- Sales analytics
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User Roles

1. Admin / Owner

Full access to the system.

Responsibilities:

- Manage users
 - Manage products
 - Manage inventory
 - Manage services
 - Manage sales
 - Manage reservations
 - View reports
 - Configure shop settings
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2. Sales Staff / Cashier

Limited access.

Responsibilities:

- Use POS
 - View inventory
 - Process sales
 - Create customer reservations
 - View reservation status
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3. Technician / Service Staff

Service-related access.

Responsibilities:

- View service appointments
 - Update repair or maintenance status
 - Record used parts
 - Add service notes
 - Save final service cost
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4. Customer

Public/user access.

Responsibilities:

- Register and log in
 - Select motorcycle type, brand, and model
 - Browse motorcycles, parts, accessories, and services
 - Reserve parts or motorcycles
 - Book service appointments
 - View reservation status
 - View estimated service cost
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Backend Modules

1. Authentication Module

Requirements:

- Customer registration
- Login and logout
- Password hashing
- Role-based access control
- Session or token authentication

- Admin and staff account management
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2. Motorcycle Profile Module

Requirements:

- Customer selects motorcycle type:
 - Underbone
 - Backbone
 - Scooter
- Customer selects brand
- Customer selects model
- Customer enters engine displacement/cc
- System uses motorcycle information to suggest compatible services and required materials

Example:

- Type: Scooter
 - Brand: Honda
 - Model: Click
 - Engine size: 125cc
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3. Service Recommendation Module

Requirements:

- Recommend services based on motorcycle type.
- Hide incompatible services.

Service rules:

- Scooter:
 - CVT Cleaning

- Change Oil
 - General Maintenance
- Underbone:
 - Chain Cleaning
 - Sprocket Cleaning
 - Change Oil
 - General Maintenance
- Backbone:
 - Chain Cleaning
 - Sprocket Cleaning
 - Change Oil
 - General Maintenance

System must not show CVT Cleaning for underbone or backbone motorcycles.

4. Service Cost Estimation Module

Requirements:

- Calculate estimated service cost based on:
 - Selected service
 - Motorcycle cc
 - Required materials
 - Labor fee
 - Product price

Change Oil Rule:

- 110cc to 125cc: 0.8 liter engine oil
- 150cc and above: 1 liter engine oil
- Labor fee: ₱50

- Total cost = oil price + labor fee

CVT Cleaning Rule:

- 125cc below: 2 cans of CVT cleaner
 - 150cc and above: 3 cans of CVT cleaner
 - Labor fee: ₱300
 - Total cost = CVT cleaner price + labor fee
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5. Inventory Module

Requirements:

- Add, edit, and delete motorcycles, parts, and accessories
 - Track stock quantity in real time
 - Set reorder level
 - Generate low-stock alerts
 - Record stock-in and stock-out
 - Prevent sales if stock is insufficient
 - Deduct used parts after sales or services
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6. Product Module

Product fields:

- Product name
- Category
- Type:
 - Motorcycle
 - Part
 - Accessory
 - Service material

- Brand
 - Model compatibility
 - Description
 - Price
 - Quantity
 - Image
 - Status:
 - Available
 - Low Stock
 - Out of Stock
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7. POS / Sales Module

Requirements:

- Search product
 - Scan product
 - Add item to cart
 - Calculate subtotal and total
 - Process payment
 - Generate receipt
 - Deduct inventory after sale
 - Save sale transaction
 - Generate sales reports
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8. Reservation Module

Reservation types:

- Motorcycle reservation

- Parts pickup reservation
- Service appointment reservation

Requirements:

- Customer chooses item or service
- Customer selects date and time
- System checks availability
- Staff receives reservation request
- Staff can approve, reject, or update reservation
- Customer can view reservation status

Reservation statuses:

- Pending
- Approved
- Rejected
- Cancelled
- Completed

9. Service Appointment Module

Requirements:

- Customer selects motorcycle type, brand, model, and engine cc
- Customer selects service type
- System suggests compatible services
- System suggests required materials
- System calculates estimated cost
- System assigns appointment slot
- Technician updates service status
- Used parts are recorded

- Final service cost is saved

Service statuses:

- Pending
- In Progress
- Completed
- Cancelled

10. Reporting Module

Reports required:

- Daily sales
- Monthly sales
- Low-stock report
- Inventory report
- Reservation history
- Service appointment report
- Best-selling products
- Inventory turnover
- Service income report